
CLEAN FEATURE ANCHORING: MITIGATING THE ACCURACY-ROBUSTNESS TRADE-OFF BY SELF-DISTILLING A NATURALLY TRAINED MODEL

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ABSTRACT

Adversarial training significantly improves adversarial robustness, but robust models attain substantially lower standard accuracy than naturally trained ones. This accuracy gap has motivated a line of work that brings a naturally trained network into adversarial training. Existing methods distill its logits into the robust model or add its predictions as an auxiliary regularizer beside the label loss. We also employ a naturally trained network, but we exploit it far more directly: the label loss is discarded entirely and the backbone is trained against the natural network alone. In this paper, we propose Clean Feature Anchoring (CFA), which anchors the adversarial feature of the student to the clean feature of the teacher rather than to its logits. We prove that this single objective bounds both fidelity to the teacher and local stability of the student up to a constant factor, so that no balancing coefficient is required. We further assign the attack radius per sample according to the input sensitivity of the loss under a fixed total budget, approximately equalizing the first-order loss change across examples. Experimental results show that CFA recovers standard accuracy without sacrificing robustness. Particularly, CFA attains 64.75% standard accuracy at 27.98% AutoAttack accuracy on CIFAR-100 with the ResNet-18 architecture, and consistent improvements are observed on CIFAR-10 and Tiny-ImageNet.

1 INTRODUCTION

Adversarial training remains the most reliable defense against norm-bounded attacks (Madry et al., 2017; Athalye et al., 2018; Croce & Hein, 2020), but its robustness comes with lower accuracy on clean inputs. This trade-off is partly intrinsic to the training objective (Tsipras et al., 2018; Zhang et al., 2019) and becomes especially restrictive for compact networks. Existing approaches improve the target seen during adversarial training: some replace one-hot labels with softened or self-distilled targets (Pang et al., 2020; Tack et al., 2022; Wu et al., 2024), while adversarial distillation transfers predictions from a robust teacher (Goldblum et al., 2020; Zhu et al., 2021; Zi et al., 2021; Huang et al., 2023; Lee et al., 2025). The latter can be effective, but presupposes a model that has already paid the cost of robust training.

A naturally trained copy of the student provides inexpensive clean guidance, despite being non-robust itself. Previous methods exploit its predictions or representations (Cui et al., 2021; Suzuki et al., 2023; Cho & Kim, 2026; Zhao et al., 2024), but choosing a target leaves two further decisions: which natural checkpoint to use, and how strongly to perturb each training example. Neither follows directly from the teacher’s clean accuracy. We study these decisions through a simple feature-regression objective, asking how the teacher and the perturbation budget shape the student’s clean-robust trade-off.

Clean Feature Anchoring (CFA) uses the natural teacher’s clean feature as a fixed target for both adversarial-example generation and backbone learning, while retaining its classifier. This gives a single objective without a label-loss coefficient or temperature search during adversarial training. A fixed-anchor bound characterizes its joint control of teacher fidelity and local feature variation. We use this simple objective to examine teacher selection and perturbation allocation.

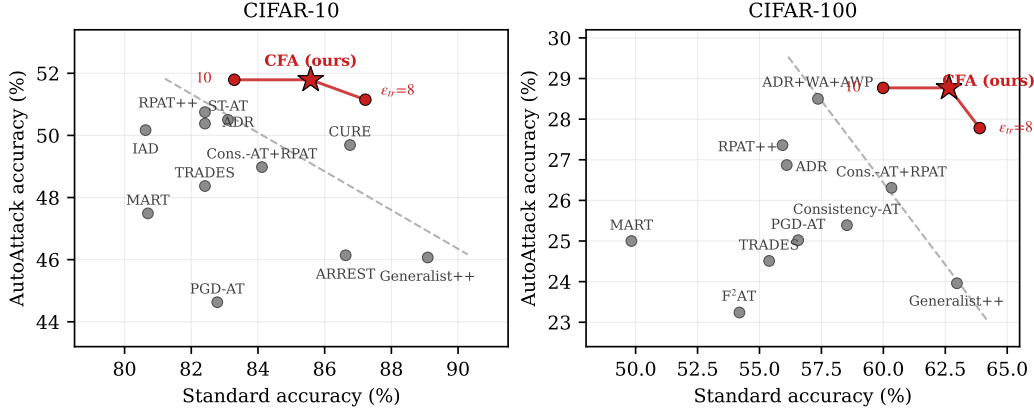


Figure 1: Clean–AutoAttack trade-off on CIFAR-10 and CIFAR-100 with ResNet-18 at $\epsilon = 8/255$. Grey points are published results (Table 28); the CFA curve varies the training radius over 100-epoch runs, and the star marks $\epsilon_{\text{train}} = 8.8/255$.

With this objective fixed, a more accurate natural teacher need not produce a more accurate student on clean inputs. Along the CIFAR-100 teacher trajectory we examine, teacher clean accuracy increases by 2.51 points, yet student clean accuracy falls by 2.04 while robustness improves. An earlier-versus-later checkpoint comparison on Tiny-ImageNet follows the same direction. Thus teacher selection changes the student’s trade-off rather than simply improving both axes. The practical implication is to assess a teacher by the student it produces, not by its own clean accuracy alone.

For a chosen teacher, a second mismatch remains: equal pixel-space radii can induce different changes in the anchor loss across examples. We therefore allocate the training radius using the input sensitivity of that same loss. Locally sensitive examples receive smaller radii and less sensitive examples larger ones, with the batch total held fixed. An inverse-sensitivity rule is motivated by equalizing the linearized loss increase. Importantly, we test the allocation signal by reassigning exactly the same radius multipliers across examples, separating their assignment from their distribution. Teacher selection and radius allocation thus address complementary choices: the reference representation and the neighborhoods over which it is preserved.

CFA achieves competitive clean–robust trade-offs across CIFAR-10, CIFAR-100, and Tiny-ImageNet. On CIFAR-100 with ResNet-18 it reaches 64.75% clean and 27.98% AutoAttack accuracy (Table 2). Controls on label supervision, regression targets, optimization recipes, and radius allocation distinguish the contribution of the anchor from that of the training stack.

Our contributions are threefold:

- We identify a teacher-selection trade-off: along the studied natural-training trajectory, higher teacher clean accuracy accompanies lower student clean accuracy and higher robustness, with the same direction observed in a Tiny-ImageNet checkpoint comparison.
- We develop sensitivity-based radius allocation for a single clean-feature objective, motivated by a fixed-budget first-order analysis and tested against controls that preserve the radius distribution.
- We evaluate CFA across three datasets and two architectures, and use target, recipe, and allocation controls to characterize where this simple natural-teacher training design is effective.

2 RELATED WORK

Adversarial training and its targets. PGD adversarial training optimizes a model on worst-case inputs (Madry et al., 2017); TRADES separates clean prediction from local consistency (Zhang et al., 2019), and MART reweights misclassified examples (Wang et al., 2020). Subsequent work improves generalization through label smoothing, consistency regularization, weight averaging, and adversarial weight perturbation (Pang et al., 2020; Tack et al., 2022; Rice et al., 2020; Wu et al., 2020). ADR replaces one-hot labels with targets produced by an annealed momentum teacher (Wu et al., 2024). These approaches motivate our study of how the supervisory target affects the clean-robust trade-off.

Adversarial distillation. ARD transfers a robust teacher’s predictions during adversarial training (Goldblum et al., 2020). RSLAD emphasizes robust soft labels (Zi et al., 2021), IAD discounts unreliable teacher guidance (Zhu et al., 2021), and AdaAD uses the teacher in both adversarial-example generation and student optimization (Huang et al., 2023). IGDM adds the teacher’s local input geometry to logit-based distillation (Lee et al., 2025). These methods are designed to inherit robustness from an adversarially trained teacher. In contrast, our teacher is trained only on clean data, and its clean representation remains fixed during the student’s inner maximization.

Guidance from naturally trained models. BGAT regresses adversarial logits to a fixed natural teacher’s clean logits, and LBGAT extends this to a jointly trained natural branch (Cui et al., 2021). ARREST starts from a naturally pretrained model and regularizes latent representations while retaining adversarial cross-entropy and noisy replay (Suzuki et al., 2023). HAT uses a natural helper to label examples outside the training ball (Rade & Moosavi-Dezfooli, 2022), DP-FAT distills direction-preserving logits (Cho & Kim, 2026), and B-MTARD combines clean and robust teachers with adaptive temperature and loss balancing (Zhao et al., 2024). CFA uses a fixed clean feature target for both attack generation and backbone learning, with an inherited classifier. We study how the natural-teacher checkpoint and sensitivity-based radius allocation affect transfer under this objective.

Sample-dependent perturbation budgets. IAAT and CAT adapt the training radius to sample difficulty, while MMA directly increases an input-space margin (Balaji et al., 2019; Cheng et al., 2022; Ding et al., 2020). Our allocation has a different source: it redistributes a fixed batch budget according to the input sensitivity of the loss being optimized. The comparison is therefore not between adaptive and uniform radii alone, but between the signals used to assign the same budget.

3 ANALYSIS

A natural teacher’s clean accuracy need not indicate how well it will guide a robust student. We first establish a simple regression target and characterize what it controls, then examine how teacher selection changes the student’s clean-robust trade-off. This separates the choice of supervisory target from the choice of teacher that supplies it.

3.1 PRELIMINARIES

Let $x \in \mathcal{X} = [0, 1]^D$ and $B(x, \epsilon) = \{x' \in \mathcal{X} : \|x' - x\|_\infty \leq \epsilon\}$. We decompose a network as $f = h \circ \Phi$, where $\Phi : \mathcal{X} \rightarrow \mathbb{R}^d$ is a continuous feature map and $h : \mathbb{R}^d \rightarrow \mathbb{R}^C$, $h(z) = Wz + b$, is the classifier for C classes. We use subscripts t and s for the frozen natural teacher and the student. Both networks share the same architecture, with fixed network statistics for the analysis; feature distances are Euclidean unless stated otherwise.

3.2 DO NATURAL TEACHERS NEED SOFTENED TARGETS?

Can the natural teacher’s concentrated clean predictions provide effective adversarial supervision if softened? Motivated by soft-label robust distillation (Goldblum et al., 2020; Zi et al., 2021; Wu et al., 2024; Zhao et al., 2024), we first vary target concentration before changing the representation

being transferred. Let

$$q_t^\tau(x) = \text{softmax}(f_t(x)/\tau), \quad p_s^\tau(x') = \text{softmax}(f_s(x')/\tau). \quad (1)$$

We compare softening only the teacher target with conventional shared-temperature KD:

$$D_{\text{KL}}(q_t^\tau(x) \| p_s^1(x')), \quad \tau^2 D_{\text{KL}}(q_t^\tau(x) \| p_s^\tau(x')). \quad (2)$$

As $\tau \rightarrow \infty$, the teacher target approaches a uniform distribution in both cases, but the scaled shared-temperature loss retains logit information. For fixed logits and $\bar{f} = f - C^{-1} \mathbf{1} \mathbf{1}^\top f$,

$$\lim_{\tau \rightarrow \infty} \tau^2 D_{\text{KL}}(q_t^\tau(x) \| p_s^\tau(x')) = \frac{1}{2C} \|\bar{f}_t(x) - \bar{f}_s(x')\|_2^2. \quad (3)$$

This limit motivates direct regression onto clean teacher logits; we also examine regression onto the full clean feature, which defines

$$\mathcal{L}_{\text{anchor}}(x) = \max_{x' \in B(x, \epsilon)} \|\Phi_s(x') - \Phi_t(x)\|_2^2. \quad (4)$$

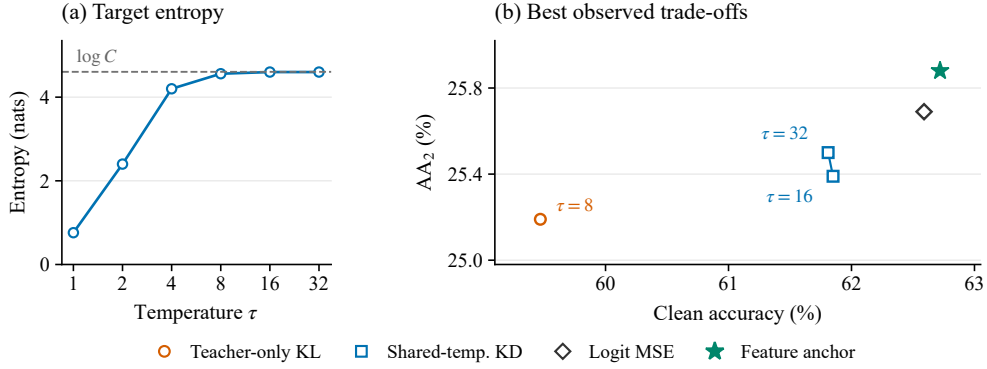


Figure 2: Target comparison on CIFAR-100 (ResNet-18). (a) Teacher-target entropy approaches the uniform value as τ increases. (b) Nondominated settings within each KD variant, alongside fixed-head logit and feature regression. AA_2 combines APGD-CE and APGD-T. The full temperature sweep is reported in Table 10.

We train for 50 epochs at $\epsilon = 8/255$, without weight averaging or AWP. Across the full sweep, teacher-only softening improves robustness until the target becomes nearly uniform, but changes clean accuracy little. Shared-temperature KD avoids this turnover and approaches logit regression (Figure 2). Feature regression is only 0.13 clean and 0.19 AA_2 points higher: both regression targets are effective without temperature selection, but this does not establish feature superiority. Moreover, KD updates the classifier whereas regression uses a fixed head, so this comparison evaluates training designs rather than isolating the target alone.

These results support direct regression to clean teacher targets in either space. We use features to preserve the representation supplied to the inherited classifier, including directions that logit regression need not constrain, and next examine what this single target controls.

3.3 WHAT THE ANCHOR CONTROLS

The anchor provides a common reference for clean fidelity and local variation. Let $x \in \mathcal{X}$ and $B = B(x, \epsilon)$, and define

$$L = \max_{x' \in B} \|\Phi_s(x') - \Phi_t(x)\|_2, \quad F = \|\Phi_s(x) - \Phi_t(x)\|_2, \quad O = \max_{x', x'' \in B} \|\Phi_s(x') - \Phi_s(x'')\|_2. \quad (5)$$

Proposition 1 (Fixed-anchor decomposition). *For every $x \in \mathcal{X}$ and $\epsilon \geq 0$,*

$$L \leq F + O \leq 3L. \quad (6)$$

Proof. For any $x' \in B$, the triangle inequality gives $\|\Phi_s(x') - \Phi_t(x)\|_2 \leq \|\Phi_s(x') - \Phi_s(x)\|_2 + F \leq O + F$; maximizing over x' yields $L \leq F + O$. Moreover, $x \in B$ implies $F \leq L$, and, for any $x', x'' \in B$, applying the triangle inequality through $\Phi_t(x)$ gives $\|\Phi_s(x') - \Phi_s(x'')\|_2 \leq 2L$. Hence $O \leq 2L$ and $F + O \leq 3L$. $\square$

If $L \leq \delta$, then the clean student feature is within δ of the teacher and the student’s feature diameter over the ball is at most 2δ , without assuming that the teacher itself is stable. The inherited linear head also bounds logit deviation by $\|W_t\|_{2 \rightarrow 2}L$, although classification depends on the teacher’s clean margin. The same argument applies to norm-based logit regression: it explains joint control by a fixed target, not a feature-specific advantage. Finite-step PGD does not certify L .

3.4 DOES TEACHER ACCURACY PREDICT TRANSFER?

The bound above describes training against a given target, but does not identify which teacher to choose. A natural criterion is the teacher’s clean accuracy. To test it, we vary the checkpoint along one natural-training trajectory while keeping the student training procedure fixed. Each checkpoint supplies the initialization, feature targets, and inherited classifier; the experiment therefore evaluates teacher selection as a whole, rather than changing the targets alone.

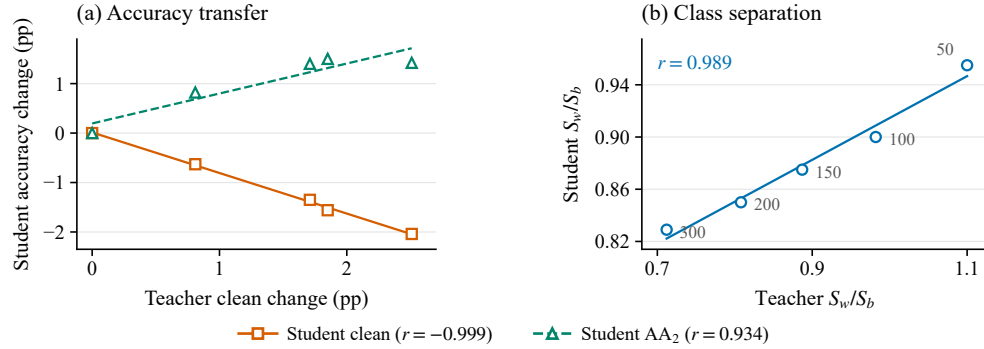


Figure 3: Teacher checkpoint sweep on CIFAR-100 (ResNet-18). (a) Accuracy changes relative to the 50-epoch teacher. (b) Within-to-between scatter ratios S_w/S_b ; lower means stronger class separation, and labels denote teacher epochs. Lines are least-squares fits. Exact values are in Table 11.

Across the sweep, the teacher gains 2.51 clean points while the student loses 2.04; student AA_2 nevertheless rises from 24.38% to 25.80% (Figure 3a). The more accurate teacher thus shifts the student toward robustness rather than improving both accuracies. Selecting the checkpoint with the highest teacher clean accuracy would miss this trade-off.

To examine what changes alongside accuracy, we measure S_w/S_b on unit-normalized clean features: S_w is the mean squared distance to the corresponding class mean, and S_b is the mean squared distance of class means to the global mean. Teacher and student ratios decrease together (Figure 3b), indicating stronger class separation. This is an association across five checkpoints, not evidence that geometry alone causes the transfer effect.

Correlations along one trajectory cannot separate a teacher property from training time, so we test the criterion directly on teachers trained differently. Label smoothing and mixup give two natural teachers whose clean accuracies differ by 0.14 points and whose students differ by 4.18 (Table 1). Their AutoAttack accuracies agree to 0.02: at matched teacher accuracy the clean axis moves and the robust axis does not. Of the teacher-only quantities we measured, the one that orders the pair is the feature sensitivity $\|\Phi_t(x + \delta) - \Phi_t(x)\|_2 / \|\delta\|_2$, which uses neither labels nor logits; a line fit on the nine trajectory checkpoints predicts 59.0 and 65.0 from it, against 64.3 and 64.2 from teacher clean accuracy, which cannot tell the two teachers apart.

The rule does not extend to every teacher. The weight-decay teacher has the sensitivity of the 200-epoch one and gives a student 7.42 clean points below it, so sensitivity is a candidate explanation for differences among teachers of comparable accuracy rather than a selector. The margin and

Table 1: Two teachers at the same clean accuracy, and a third that refutes the general rule. CIFAR-100, ResNet-18, 50-epoch students identical apart from the teacher. Sensitivity is the feature displacement per unit input displacement, measured on held-out data.

Natural training	Teacher		Student	
	Clean	Sensitivity	Clean	AA ₂
200 epochs (Figure 3)	77.38	4.14	62.72	25.88
+ label smoothing 0.1	78.76	3.17	59.62	25.72
+ mixup	78.62	4.81	63.80	25.70
+ weight decay 5×10^{-3}	69.30	4.13	55.30	19.60

separation metrics, which correlate most strongly along the trajectory, fail outside it; Appendix G reports those predictions, the full metric set and the retracted checkpoint. What survives is that teachers of equal accuracy transfer differently, so checkpoints have to be assessed by the student trade-off they produce. Our subsequent experiments use the 200-epoch teacher, which lies on the late-checkpoint robustness plateau while retaining more student clean accuracy than the 300-epoch teacher. Once the teacher is fixed, a separate question remains: how should the perturbation budget be distributed across examples whose anchor losses have different input sensitivities? We address this next.

4 METHOD

CFA combines a clean-feature training objective with sample-dependent perturbation budgets. Teacher selection fixes the reference representation; radius allocation determines the neighborhood over which each example is trained to preserve it.

4.1 CLEAN FEATURE ANCHORING

Let Φ_t and h_t be the frozen feature map and classifier of the natural teacher. The student inherits both, $\Phi_s \leftarrow \Phi_t$ and $h_s = h_t$, after which only Φ_s is optimized. For a clean example x , CFA solves

$$x_{\text{adv}} = \arg \max_{x' \in B(x, \epsilon)} \|\Phi_s(x') - \Phi_t(x)\|_2^2, \quad (7)$$

$$\mathcal{L}_{\text{CFA}} = \mathbb{E}_x [\|\Phi_s(x_{\text{adv}}) - \Phi_t(x)\|_2^2].$$

We approximate the inner maximization with randomly initialized PGD against the fixed target $\Phi_t(x)$ and minimize the same discrepancy over the backbone. This adversarial-training stage uses no label loss; inference uses $h_t \circ \Phi_s$. Algorithm 1 gives the complete procedure.

4.2 SENSITIVITY-MATCHED ϵ

A uniform pixel radius fixes the size of each input neighborhood, but not the change it induces in the anchor loss. An example whose loss is locally sensitive can experience a larger increase than another example given the same radius. We use this discrepancy to allocate a fixed batch budget, rather than increasing the mean radius or adding another loss term. For $\ell_{\text{anc}}(x'; x_i) = \|\Phi_s(x') - \Phi_t(x_i)\|_2^2$, define

$$g_i = \|\nabla_{x'} \ell_{\text{anc}}(x'; x_i)\|_1|_{x'=x_i}, \quad \max_{\|\delta\|_\infty \leq \epsilon} \langle \nabla_{x'} \ell_{\text{anc}}, \delta \rangle = \epsilon g_i. \quad (8)$$

The linearized worst-case loss increase under radius ϵ_i is therefore $\epsilon_i g_i$. To keep this local increase comparable across examples, a more sensitive example should receive a smaller radius. For $g_i > 0$, equalizing the products subject to $\sum_i \epsilon_i = N\epsilon$ gives

$$\epsilon_i = N\epsilon \frac{g_i^{-1}}{\sum_j g_j^{-1}}, \quad \text{or more generally} \quad \epsilon_i \propto g_i^{-p}, \quad (9)$$

where $p = 0$ recovers a uniform radius and $p = 1$ gives inverse-sensitivity allocation. The conserved sum separates redistribution from a change in the mean training budget. This is a local approximation to loss growth, not a claim that nonlinear adversarial losses become equal.

We use $p = 1$ with an ℓ_2 gradient norm as a practical sensitivity proxy, measured with one backward pass at clean inputs. The ℓ_1 norm follows exactly from the linearized ℓ_∞ problem; it also retains the clean-accuracy gain but gives 0.29 points lower AA in our comparison (Appendix E). Thus first-order equalization motivates the allocation, while the implemented proxy is chosen empirically.

We floor the proxy at 10^{-12} , clip its inverse-mean multipliers to $[0.5, 1.5]$, and normalize them to unit mean before setting ϵ_i . The floor gives a uniform allocation when all clean gradients vanish at teacher initialization. Normalization preserves $\sum_i \epsilon_i = N\epsilon$, although the final multipliers can leave the clipping interval. PGD step sizes are scaled by the same multipliers. Appendix E specifies this procedure and an optional exact-box variant. The key test is whether sensitivity assigns the budget usefully, not merely whether varying radii helps. Accordingly, Table 6 includes a control that keeps the exact multiset of sensitivity-derived multipliers but reassigns them by sample difficulty. Together with uniform and margin-based controls, this distinguishes the allocation signal from the spread of radii.

5 EXPERIMENTS

We evaluate the clean-robust trade-off across datasets and architectures, isolate the anchor’s contribution, and distinguish radius-allocation effects from weight averaging and AWP. We then examine feature geometry and attack reliability.

5.1 EXPERIMENTAL SETTINGS

We use CIFAR-10, CIFAR-100 (Krizhevsky et al., 2009), and Tiny-ImageNet-200 (Le & Yang, 2015). The natural teacher is trained on clean data for 200 epochs and then frozen (Table 21); its clean accuracies are 95.32%, 77.66%, and 66.29%, respectively, and its AA is zero on all three datasets. Unless stated otherwise, the ResNet-18 student is initialized from that teacher and trained with AdamW, a one-cycle schedule, and 10-step PGD, with sensitivity matching ($p = 1$), weight averaging, and AWP. The main CIFAR results use 150 epochs at the standard $\epsilon_{\text{train}} = 8/255$; ablations use 100 epochs and, where stated, $\epsilon_{\text{train}} = 8.8/255$, and Table 16 sweeps the radius. Baselines run their authors’ recipes. Detailed settings are in Appendix F.

We evaluate the final-epoch model on the full test set at $\epsilon = 8/255$, reporting clean accuracy, the APGD-CE/APGD-T cascade (AA), their mean (Avg), and their harmonic mean (NRR); a four-attack AutoAttack validation of the 100-epoch CIFAR-100 model gives the same robust accuracy (Appendix I). Other baseline settings follow their published recipes with our evaluation pipeline, and the distillation baselines receive our natural teacher; published results obtained with other teachers are collected in Table 28.

5.2 MAIN RESULTS

CFA attains the highest AA and NRR on both CIFAR datasets (Table 2). On CIFAR-100 it improves over ADR, the closest baseline in NRR, by 0.78 AA and 1.65 NRR while gaining 4.82 clean points, and it exceeds the closest CIFAR-10 baseline by 1.85 NRR. ARREST reaches the highest clean accuracy of the table but gives up 5.20 AA points, placing it elsewhere on the trade-off.

5.3 GENERALIZATION ACROSS DATA AND MODEL SCALE

On Tiny-ImageNet, CFA attains the highest Avg and NRR while maximizing neither endpoint: ARREST stays higher on clean accuracy and ADR is 0.31 points higher on AA (Table 3). The teacher-checkpoint effect transfers here too, since the 80-epoch teacher moves the student from 55.16/20.54 to 57.08/18.96 clean/AA. On CIFAR-10 with WideResNet-34-10 CFA is highest on both axes, matching ADR’s clean accuracy (88.67% versus 88.64%) with 2.95 points more AA; CIFAR-100 on that network is in Table 20, where CFA again leads ADR on both.

5.4 WHAT PRODUCES THE GAIN?

Anchor and label supervision. Replacing the anchor by label cross-entropy under the same base regime reduces clean and AA by 6.69 and 5.91 points, and giving cross-entropy its own optimizer

Table 2: ResNet-18 results at ℓ_∞ , $\epsilon = 8/255$. All values are measured in our framework. T denotes the teacher assumed by the method: none (–), a natural model (nat.), or the method’s own weight average (self).

Method	T.	CIFAR-10				CIFAR-100			
		Clean	AA	Avg	NRR	Clean	AA	Avg	NRR
PGD-AT	–	84.54	43.57	64.06	57.50	57.36	20.34	38.85	30.03
TRADES	–	81.71	49.16	65.44	61.39	55.28	23.55	39.42	33.03
MART	–	80.41	45.56	62.99	58.16	53.86	22.72	38.29	31.96
ARD	nat.	84.58	43.75	64.16	57.67	57.61	20.24	38.93	29.96
RSLAD	nat.	85.15	42.91	64.03	57.06	59.68	21.30	40.49	31.40
AdaAD	nat.	85.39	46.22	65.81	59.98	59.79	23.19	41.49	33.42
AdaAD + IGDM	nat.	85.01	45.50	65.26	59.27	48.25	19.48	33.87	27.75
LBGAT	nat.	–	–	–	–	56.46	26.02	41.24	35.62
ARREST	nat.	85.06	46.42	65.74	60.06	66.97	22.78	44.88	34.00
HAT	nat.	84.59	48.81	66.70	61.90	60.67	24.45	42.56	34.85
ADR	self	83.36	49.72	66.54	62.29	59.93	27.20	43.57	37.42
RPAT++	–	82.41	50.75	66.58	62.82	55.93	27.36	41.64	36.74
CFA (ours)	nat.	87.37	51.33	69.35	64.67	64.75	27.98	46.37	39.07

Table 3: A larger dataset and a larger network, both measured in our framework: Tiny-ImageNet-200 with ResNet-18, and CIFAR-10 with WideResNet-34-10. Table 20 reports WideResNet-34-10 on CIFAR-100.

Method	T.	Tiny-ImageNet, ResNet-18				CIFAR-10, WRN-34-10			
		Clean	AA	Avg	NRR	Clean	AA	Avg	NRR
PGD-AT	–	49.72	14.45	32.09	22.39	86.00	45.71	65.86	59.69
TRADES	–	46.98	15.95	31.46	23.83	85.13	48.99	67.06	62.19
MART	–	46.01	16.90	31.46	24.73	85.29	47.06	66.18	60.65
ARD	nat.	49.74	14.48	32.11	22.43	86.35	45.52	65.94	59.61
RSLAD	nat.	51.26	17.02	34.14	25.56	86.41	45.80	66.10	59.87
AdaAD	nat.	51.59	16.79	34.19	25.33	86.80	46.84	66.82	60.85
LBGAT	nat.	51.04	18.47	34.75	27.13	81.56	51.79	66.68	63.35
ARREST	nat.	59.38	16.06	37.72	25.28	87.54	48.83	68.19	62.69
HAT	nat.	52.57	17.28	34.93	26.01	87.15	53.21	70.18	66.07
ADR	self	47.66	20.85	34.25	29.01	88.64	52.34	70.49	65.82
CFA (ours)	nat.	55.16	20.54	37.85	29.93	88.67	55.29	71.98	68.11

and schedule narrows the gap without closing it, leaving CFA ahead by 4.63 clean and 4.45 AA points (Table 4).

Table 4: Objective control on CIFAR-100 with ResNet-18, 100 epochs, no weight averaging or AWP, $p = 0$, and $\epsilon_{\text{train}} = 8.8/255$.

Objective	Clean	AA	NRR
Anchor	61.29	25.36	35.88
Label cross-entropy, same recipe	54.60	19.45	28.68
Label cross-entropy, own recipe	56.66	20.91	30.55

Changing initialization or optimizer and schedule changes NRR by 0.22 and 0.11, respectively (Table 25). At 100 epochs with the full stack, label cross-entropy reaches 57.73/26.46 clean/AA versus CFA’s 62.17/28.86. Giving each baseline the same warm start and stack at its own recipe raises most of them, PGD-AT by 6.00 NRR and ARD by 6.45, and none reaches CFA (Table 5).

Regression and probability targets. Logit regression remains effective with WA and AWP: at a uniform $8.8/255$ and 100 epochs, logit MSE reaches 60.36/28.02 clean/AA against feature regression’s 60.42/28.42 (Table 12), so with the base-regime comparison of Section 3.2, effective clean-target regression is not confined to feature space. Teacher-only KL reaches 56.97/26.96 at its

best tested temperature under the full recipe (Table 27), a comparison that does not isolate the target from classifier and radius choices.

Table 5: Baselines given the natural warm start and the training stack, CIFAR-100 with ResNet-18. Each method keeps its own objective, optimizer, schedule and epochs; the right block adds teacher initialization, weight averaging and the AWP proxy, the latter on that method’s own loss. Blank entries are in progress. [†]ADR’s own recipe already carries its own SAM-style AWP, so its left column is not stack-free; without that term it gives 58.22/20.93 (NRR 30.79).

Method	Own recipe			+ teacher init + WA + AWP		
	Clean	AA	NRR	Clean	AA	NRR
PGD-AT	57.36	20.34	30.03	59.85	25.77	36.03
TRADES	55.28	23.55	33.03	56.16	24.56	34.17
MART	53.86	22.72	31.96	50.97	25.13	33.66
ARD	57.61	20.24	29.96	60.19	26.10	36.41
RSLAD	59.68	21.30	31.40	63.44	25.55	36.43
AdaAD	59.79	23.19	33.42	57.21	27.10	36.78
AdaAD + IGDM	48.25	19.48	27.75	54.24	25.56	34.75
Consistency-AT	57.53	20.01	29.69	60.74	25.38	35.80
LBGAT	56.46	26.02	35.62	53.91	25.16	34.31
ADR [†]	59.93	27.20	37.42	59.84	27.84	38.00
HAT	60.67	24.45	34.85	57.63	24.74	34.62
ARREST	66.44	22.99	34.16	64.95	23.02	33.99
CFA (ours)	—			64.75	27.98	39.07

The stack helps nine of the twelve, most where the objective was weakest: AdaAD + IGDM gains 7.04 NRR, ARD 6.45, PGD-AT 6.00, Consistency-AT 6.11. It does not help everywhere — LBGAT loses 1.31, HAT 0.23 and ARREST 0.17, the last two on schedules too short for robust overfitting to undo — and no stacked baseline reaches CFA, the closest being ADR at 1.07 NRR behind.

5.5 RADIUS ALLOCATION AND TRAINING COMPONENTS

Which signal should allocate the radius? Table 6 holds the mean training radius fixed and changes the allocation signal under the full recipe, where sensitivity improves clean/AA by 1.75/0.44 over a uniform radius. Preserving the rule’s exact multipliers but reassigning them so easier examples receive larger radii costs 0.80 AA, and difficulty and margin signals also have lower NRR, which supports allocating from the anchor’s own input sensitivity. These are signal controls inspired by adaptive-radius methods rather than reproductions of those algorithms.

Table 6: Radius-allocation signals under the same mean radius, CIFAR-100, ResNet-18, 100 epochs with WA and AWP. $sd(w)$ measures multiplier dispersion.

Allocation	$sd(w)$	Clean	AA	NRR
Uniform	0	60.42	28.42	38.66
Sensitivity	0.32	62.17	28.86	39.42
Same multipliers, difficulty order	0.32	60.90	28.06	38.42
Difficulty magnitude	0.39	61.27	28.12	38.55
Logit margin	0.42	61.07	27.79	38.20

Adding weight averaging and AWP. Table 7 follows the additions to the base anchor at 100 epochs: sensitivity matching raises clean/AA by 1.80/0.38, weight averaging adds 1.95 AA points and AWP 1.17, a total NRR gain of 3.54. These are incremental effects in the stated order and do not isolate interactions between components; radius sweeps and WA/AWP comparisons on both CIFAR datasets are in Tables 16 and 17.

Feature geometry. Additional base-regime diagnostics show stronger class separation under anchoring, but not smaller relative feature displacement under a matched attack (Appendix C.2). These earlier trained-head models provide descriptive evidence, not a causal account of the final method.

Table 7: Component ablation on CIFAR-100 with ResNet-18, 100 epochs, and $\epsilon_{\text{train}} = 8.8/255$. Each row adds one component to the row above.

	Clean	AA	NRR
Anchor ($p = 0$)	61.29	25.36	35.88
+ sensitivity-matched ϵ	63.09	25.74	36.56
+ weight averaging	62.30	27.69	38.34
+ AWP	62.17	28.86	39.42

5.6 RELIABILITY AND SCOPE

The full AutoAttack suite (APGD-CE, APGD-T, FAB-T, Square) reaches 28.73% on the 100-epoch CIFAR-100 repeat, identical to the two-attack cascade, and PGD accuracy falls to zero at 64/255 (Appendix I). That repeat gives 62.00/28.74 clean/AA against 62.17/28.86 for seed 0, so two seeds do not establish the significance of small ablation differences (Appendix F). Teacher and student training take 0.39 and 2.07 hours on one RTX 5080 (Table 22).

6 CONCLUSION

Effective transfer from a natural teacher depends on more than its clean accuracy: in the checkpoint comparisons studied here, a more accurate teacher can give a less accurate student while improving robustness. CFA makes that trade-off easy to examine, since one fixed clean feature target drives both attack generation and backbone learning, and the sensitivity-matched radius then matches the perturbation budget to the loss being optimized. Across three datasets the result is competitive with neither a robust teacher nor a label objective, which argues for choosing teachers by the student trade-off they produce.

A ALGORITHM

Algorithm 1 Clean Feature Anchoring (CFA)

```

1: Input: dataset  $\mathcal{D}$ , radius  $\epsilon$ , attack steps  $K$ , step size  $a$ , exponent  $p$ , clipping thresholds  $[w_{lo}, w_{hi}]$ 
2: Output: robust student  $h_t \circ \Phi_s$ 
3:
4: {Phase 1 — natural self-teacher. The only stage that uses labels.}
5: Train  $f_t = h_t \circ \Phi_t$  on  $\mathcal{D}$  with standard cross-entropy; freeze it
6:
7: {Phase 2 — anchored adversarial training. No labels, no head update.}
8:  $\Phi_s \leftarrow \Phi_t$  {student initialized at the teacher}
9: for epoch = 1 to  $T$  do
10:   for minibatch  $\{x_i\}_{i=1}^N \subset \mathcal{D}$  do
11:      $\phi_i \leftarrow \Phi_t(x_i)$  {teacher read once, at the clean input}
12:      $g_i \leftarrow \max(\|\nabla_x \|\Phi_s(x_i) - \phi_i\|_2, 10^{-12})$  {one backward pass}
13:      $v_i \leftarrow \text{clip}((\bar{g}/g_i)^p, w_{lo}, w_{hi})$ ;  $w_i \leftarrow Nv_i / \sum_j v_j$ 
14:      $\epsilon_i \leftarrow \epsilon w_i$ ;  $a_i \leftarrow aw_i$  { $\sum_i \epsilon_i = N\epsilon$ }
15:      $x_i^0 \leftarrow x_i + \eta_i$ 
16:     for  $k = 0$  to  $K - 1$  do
17:        $x_i^{k+1} \leftarrow \Pi_{B(x_i, \epsilon_i)}(x_i^k + a_i \text{sign } \nabla_x \|\Phi_s(x_i^k) - \phi_i\|_2)$ 
18:     end for
19:      $\mathcal{L} \leftarrow \frac{1}{N} \sum_i \|\Phi_s(x_i^K) - \phi_i\|^2$ 
20:      $\Phi_s \leftarrow \Phi_s - \mu \nabla_{\Phi_s} \mathcal{L}$  { $h_t$  is never updated}
21:   end for
22: end for

```

B TWO PARTIAL RESULTS ON THE ROBUSTNESS SIDE

Natural-teacher guidance is shared with prior work; our question is whether a fixed clean target can support adversarial training without an additional label objective. In CFA, the teacher specifies the target while the student’s inner maximization enforces matching under perturbations. Teacher vulnerability alone therefore does not determine the student’s response: matching a clean target is different from reproducing the teacher at perturbed inputs.

The two results below examine this distinction in restricted settings. The first studies robust and non-robust features in a two-feature model; the second relates anchor error to a classification margin. They supplement the fixed-anchor decomposition, but do not predict the measured accuracy or establish superiority over other regression targets.

B.1 IN THE TWO-FEATURE MODEL, THE NON-ROBUST COEFFICIENT IS ZEROED RATHER THAN SHRUNK

The setting is the two-feature model of Tsipras et al. (2018), which is where the objection is usually posed. One feature x_1 agrees with the label with probability $p = 0.95$ and cannot be attacked. A further d features $z_j \sim \mathcal{N}(\eta y, 1)$ are individually almost uninformative, are jointly close to perfectly informative through their mean $m(z)$, and can all be moved by an ℓ_∞ adversary. A naturally trained network therefore reads $m(z)$: near-perfect clean accuracy, no robustness whatever. That is our teacher’s profile, so $\Phi_t = m(z)$.

Write the student’s feature as $\Phi_s = ax_1 + bm(z)$. Then b is exactly the quantity the objection is about: how much of the teacher’s fragile route the student takes over. If the objection is right, minimizing our objective should leave b large.

Proposition 2. *With the inner maximum exact, $J(a, b) = \mathbb{E}[R^2] + 2\epsilon|b| \mathbb{E}|R| + \epsilon^2 b^2$ with $R = ax_1 + (b-1)m$, kinked at $b = 0$ with subgradient jump $2\epsilon \mathbb{E}|R(a, 0)| > 0$; above a threshold ϵ_0 the minimizer has $b^* = 0$ exactly.*

Proof. The adversary reaches Φ_s only through m , and only through the mean perturbation $\bar{\delta} = \frac{1}{d} \sum_j \delta_j$, which ranges over $[-\epsilon, \epsilon]$ as $\|\delta\|_\infty \leq \epsilon$. The inner objective is therefore $\max_{|\bar{\delta}| \leq \epsilon} (R + b\bar{\delta})^2 = (|R| + \epsilon|b|)^2$, since a scalar quadratic is maximized at an endpoint and the sign of $\bar{\delta}$ is free. Taking expectations and expanding the square gives the display.

Fix a and differentiate in b . The term $\mathbb{E}[R^2]$ and $\epsilon^2 b^2$ are smooth, while $2\epsilon|b| \mathbb{E}|R|$ contributes $\pm 2\epsilon \mathbb{E}|R|$ as $b \rightarrow 0^\pm$. The one-sided derivatives at $b = 0$ are thus $2\mathbb{E}[R(a, 0) m] \pm 2\epsilon \mathbb{E}|R(a, 0)|$, so $b = 0$ is a minimum along b exactly when

$$|\mathbb{E}[R(a, 0) m]| \leq \epsilon \mathbb{E}|R(a, 0)|. \quad (10)$$

At $b = 0$ the objective in a is $\mathbb{E}(ax_1 - m)^2$, minimized at $a^* = \mathbb{E}[x_1 m] / \mathbb{E}[x_1^2] = q\eta$, and there $\mathbb{E}[R(a^*, 0) m] = q^2 \eta^2 - (\eta^2 + \frac{1}{d}) = -c$ with

$$c = \eta^2 v + \frac{1}{d}, \quad v = 1 - q^2. \quad (11)$$

Equation (10) therefore holds for every

$$\epsilon \geq \epsilon_0 = \frac{c}{\mathbb{E}|q\eta x_1 - m|}, \quad (12)$$

which is the stated threshold. The mechanism is visible in Equation (10): the left side is fixed while the right grows linearly in ϵ , so the kink's slope overtakes the quadratic gain at a finite radius rather than asymptotically. $\square$

Why the coefficient is deleted rather than shrunk. The adversary contributes $2\epsilon|b| \mathbb{E}|R|$ to the loss — linear in $|b|$, not quadratic in b . A quadratic penalty pushes a coefficient toward zero and never arrives; a penalty on $|b|$ is kinked at the origin and sets the coefficient to zero outright once its slope dominates, which is the same distinction that makes the lasso sparse and the ridge not. What produces the kink here is that the student is asked to reproduce a *value* rather than a *route*: the fragile features carry that value on clean data but not inside the ball, so past a certain radius carrying any of them costs more than it pays. The teacher supplies what to match, and the inner maximization discards how the teacher got there.

At $d = 512$ the model's constants give $c = 7.89 \times 10^{-3}$ and $\mathbb{E}|q\eta x_1 - m| = 0.0529$, hence $\epsilon_0 = 0.843 \eta$ — inside the bracket $(0.5\eta, 1.0\eta)$ that the Monte-Carlo minimization returns, and far below the 2η at which the model is usually stated.

Two caveats are worth stating rather than leaving to be found. First, Equation (12) is a condition for $b = 0$ to minimize along b at $a = a^*$; J is not jointly convex, because $|b| \mathbb{E}|R|$ is a product of two convex factors, so global minimality is what the numerical check below supplies and not what the argument proves. Second, the model assumes x_1 is unperturbable and z fully perturbable, so *some* suppression of z is built in before anything is shown. What is not built in is exactness, the threshold, and the ϵ -independence that follows it.

That the exactness is the kink's doing, and not a general consequence of adding an adversary, is visible from dropping it. With the $|b|$ term removed, $J_{\text{quad}} = \mathbb{E}[R^2] + \epsilon^2 b^2$; minimizing out a and writing $u = 1 - b$ collapses it to $u^2 c + \epsilon^2 (1 - u)^2$, so

$$b_{\text{quad}}^* = \frac{c}{c + \epsilon^2}, \quad (13)$$

which is strictly positive at every radius. Against the exact minimizer ($d = 512$, Monte-Carlo, $n = 2 \times 10^5$):

ϵ	b^* (exact)	b_{quad}^* (kink dropped)
0.5η	0.510	0.502
1.0η	0.000	0.202
2.0η	0.000	0.060
4.0η	0.000	0.016

The two columns agree below ϵ_0 and separate above it: the quadratic surrogate shrinks the non-robust coefficient and never deletes it, which is the ordinary picture, while the kinked objective sets it to zero and keeps it there.

Since every ϵ in J multiplies b , at $b^* = 0$ the objective is $\mathbb{E}(ax_1 - m)^2$ and is ϵ -free—past the threshold a larger radius costs the anchor nothing, which matches the saturation measured in Table 16, where AutoAttack and CW at 10/255 are identical to the last decimal to those at 8.8/255 on both datasets.

This model rules out the paper’s effect by construction, so Proposition 2 cannot be the explanation of it: its reliability spectrum has two atoms and no mass near $\eta_j/\epsilon = 1$, where the effect lives. It is a statement about the free half only.

B.2 WHAT ROBUST ACCURACY TRACKS IS THE LOSS MEASURED AGAINST A MARGIN

Writing $\gamma_t(x)$ for the teacher’s margin at x and D_y for the relevant decision-boundary constant, Cauchy–Schwarz gives that $\gamma_t(x) > D_y \cdot L(x)$ implies the student is robustly correct at x . The certificate is vacuous at the scales we train at, but the ratio DL/γ_t tracks both axes across the teacher ladder of Figure 3. What it does not do is predict the size of the effect, and it saturates before AutoAttack does.

C SUPPLEMENTARY ANALYSIS OF THE CLEAN-FEATURE ANCHOR

C.1 TEACHER EVALUATION ON PERTURBED INPUTS

The comparison in Section 3.2 evaluates the teacher at clean inputs. Here we examine why its responses to perturbed inputs may be unsuitable as supervisory targets. Under $\epsilon = 8/255$, the natural teacher’s feature rotates by 63.8° and its norm grows by a factor of 2.45. Its representation can therefore change substantially within the perturbation ball.

As a control, replacing the clean target $\Phi_t(x)$ with $\Phi_t(x')$ makes an exact copy of the teacher, $\Phi_s = \Phi_t$, a zero-loss solution even when the teacher is sensitive to perturbations. This variant yields 0.00% AutoAttack accuracy (Table 8). With a fixed clean target, copying the teacher instead incurs a loss wherever its features change within the ball.

Table 8: Where the teacher is evaluated. CIFAR-100, ResNet-18, 50 epochs; the two rows differ only in the teacher input used by the loss and inner maximization.

Teacher target	Clean	AA	NRR
$\Phi_t(x)$	61.33	26.19	36.71
$\Phi_t(x_{\text{adv}})$	76.11	0.00	0.00

IGDM (Lee et al., 2025) uses the finite difference $f_t(x + \delta) - f_t(x - \delta)$ to approximate the teacher’s response along the perturbation direction. Its accuracy depends on local linearity. The reported Taylor remainder proportion is 0.012 for adversarially trained teachers; we obtain 0.0111 for an adversarially trained ResNet-18 and 0.4763 for the natural teacher (Table 13). For the natural teacher, the finite difference has norm 14.14, compared with $\|f_t(x)\| = 18.36$, and its softmax assigns 0.917 probability to one class. These measurements indicate a less accurate local linear approximation and a highly concentrated target distribution.

We also evaluate published distillation objectives with the natural teacher replacing their robust teacher, using each method’s published recipe (Table 18). All evaluated objectives perform below adversarial training initialized from that teacher. This comparison complements the target sweep in Section 3.2, although differences among training recipes prevent it from isolating the target alone.

C.2 ADDITIONAL EVIDENCE FOR THE ANCHOR OBJECTIVE

The geometry and displacement diagnostics below use the earlier 100-epoch `ladder_p0_100ep` anchor and `abl_ce_nostack` checkpoints. The anchor’s backbone uses raw feature regression, while a detached KD term trains its classifier; this is distinct from the frozen-head models in Table 4. Both diagnostics use true-label PGD with 10 steps at radius 8.8/255. The displacement script samples training batches, whereas the class-geometry script samples test batches. Consequently, they are complementary descriptive measurements rather than a joint causal test on the final model.

Proposition 1 bounds variation within an anchored model, but does not predict a smaller variation than label training. Under the matched true-label PGD protocol, the anchor and label-trained students move by 0.275 and 0.254 of their clean feature norms. These displacements are not the worst-case diameter O and provide no evidence of smaller relative displacement under anchoring. Instead, the stronger difference is in class organization: the anchor has a $5.5\times$ smaller clean S_w/S_b and a $2.7\times$ smaller ratio under attack (Table 9).

Table 9: Feature geometry under the same true-label PGD attack, using the earlier base-regime models described above. Angles are measured on unit-normalized features. *Own* is the sample-to-own-centroid angle; *gap* is the mean angle from each class centroid to its nearest other-class centroid.

	Clean			Adversarial		
	S_w/S_b	Own	Gap	S_w/S_b	Own	Gap
Anchor	0.898	28.1°	28.6°	2.681	33.3°	16.2°
Label CE	4.960	45.4°	21.6°	7.143	47.0°	18.3°

The classification benefit is evaluated in Section 5.4. Without weight averaging, AWP, or the sensitivity-matched radius, the anchor improves clean and AutoAttack accuracy over label cross-entropy by 6.69 and 5.91 points under the same training conditions. The gains are 4.63 and 4.45 points when cross-entropy uses its own optimizer and schedule (Table 4, Table 25). At the full training configuration, the AutoAttack gain over cross-entropy with weight averaging and AWP is 2.40 points. Under a matched attack, the students exhibit similar relative feature displacement, while their class separation differs by a factor of 5.5. This suggests a contribution from the learned geometry, but does not establish class separation as the sole cause of the robustness gain.

The checkpoint ladder in Figure 3 further separates within-class concentration from between-class separation on unit-normalized features. From the 50-epoch to the 300-epoch teacher, the mean angle from a sample to its class centroid decreases from 34.6° to 30.9°, whereas the nearest-other-centroid angle between class means increases from 35.7° to 44.9°. The corresponding students reproduce the latter change—their nearest-centroid separation increases from 25.4° to 29.8°—while their mean within-class angle remains nearly unchanged (27.9° to 28.2°). These measurements motivate interpreting the ladder through between-class geometry, but they remain observational because the teacher’s accuracy and geometry change together. The numerical values underlying the two analysis figures are collected below.

Table 10: Numerical values for the target comparison in Figure 2. $H(q_t^\tau)$ is measured in nats, and NRR is the harmonic mean of clean and AA₂ accuracy.

Target	τ	$H(q_t^\tau)$	Clean	AA ₂	NRR
<i>Teacher-only temperature</i>					
KL	1	0.76	58.26	20.84	30.70
KL	2	2.40	59.33	22.79	32.93
KL	4	4.20	59.39	24.48	34.67
KL	8	4.56	59.47	25.19	35.39
KL	16	4.60	57.78	24.00	33.91
<i>Shared-temperature KD</i>					
KL	4	4.20	60.83	25.05	35.49
KL	8	4.56	60.99	25.25	35.71
KL	16	4.60	61.85	25.39	36.00
KL	32	4.60	61.81	25.50	36.11
Logit regression, fixed head	—	—	62.59	25.69	36.43
Feature anchor, fixed head	—	—	62.72	25.88	36.64

Table 11: Numerical values for the checkpoint trends in Figure 3.

Checkpoint	Teacher		Student		
	Clean	S_w/S_b	Clean	S_w/S_b	AA ₂
50 epochs	75.81	1.100	64.28	0.955	24.38
100 epochs	76.62	0.982	63.65	0.900	25.20
150 epochs	77.52	0.887	62.93	0.875	25.78
200 epochs	77.66	0.808	62.72	0.850	25.88
300 epochs	78.32	0.712	62.24	0.829	25.80

C.3 SCOPE OF THE ANCHOR BOUND

Proposition 1 concerns exact maxima over the perturbation set. A loss evaluated using a finite attack does not certify these maxima. The factor 3 combines $F \leq L$ with $O \leq 2L$; the latter bound is attained when two student features lie on opposite sides of the target, each at distance L .

The argument applies to any fixed target in a normed space, including logits when matched using a norm distance. It explains how one anchor controls two quantities, but does not establish that features are superior to logits. The feature choice is motivated by Section 3.2. Likewise, the bound does not directly lower-bound robust classification accuracy: this requires information about the classifier and class margins, as discussed in Appendix B.

Joint control also does not imply that fidelity and stability can both be made arbitrarily small over the dataset. When perturbation balls overlap but their teacher targets differ, they impose conflicting constraints on the same student feature. The proposition describes how the objective controls the two errors without eliminating this possible conflict.

D ADDITIONAL RESULTS

The following tables provide additional results supporting the main experiments and supplementary analysis.

Regression under weight averaging and AWP. Table 12 compares logit and feature regression with the same natural teacher, inherited frozen classifier, AdamW optimizer, one-cycle schedule, WA, and AWP. Both use a uniform radius: neither run includes sensitivity-based allocation. Each objective is maximized to generate training perturbations and minimized to update the backbone, so the comparison includes its effect on both stages. Logit regression remains competitive with the training stack; the single-seed difference does not establish a general advantage for feature regression.

Table 12: Fixed-head regression with WA and AWP: CIFAR-100, ResNet-18, 100 epochs, uniform $\epsilon_{\text{train}} = 8.8/255$. Evaluation uses the APGD-CE/APGD-T cascade at $8/255$. Both losses sum squared errors across output coordinates and average over the batch.

Target	Clean	AA
Clean logits	60.36	28.02
Clean features	60.42	28.42

Table 13: Local linearity, which IGDM’s construction requires of the teacher: remainder proportion of the first-order Taylor expansion over an $\epsilon = 8/255$ ball, computed as in IGDM’s Sec. 3.1.

Network	Remainder proportion
IGDM’s robust teachers (LTD / BDM-AT / IKL-AT, published)	0.012 / 0.012 / 0.016
PGD-AT ResNet-18 (ours)	0.0111
Natural teacher (ours)	0.4763
Anchored student, $p = 0$	0.0029
Anchored student, shipped recipe	0.0202

Table 14: What to do with the classifier. CIFAR-100 base regime (50 epochs, raw features, no weight averaging, no AWP, $\epsilon = 8/255$), so that no stack component can absorb the difference.

	Clean	PGD-20	CW	AA	NRR
Logit target only, $\tau = 1$	58.26	22.62	22.79	20.84	30.70
Logit target only, $\tau = 4$	59.39	30.30	26.84	24.48	34.67
Logit target only, $\tau = 16$	57.78	31.57	26.11	24.00	33.91
Feature anchor + head KD, $\tau = 1$	62.34	26.34	26.11	23.93	34.58
Feature anchor + head KD, $\tau = 16$	62.61	32.47	27.32	25.61	36.35
Feature anchor, head untouched	62.72	28.69	27.80	25.88	36.64
head refit, adversarial CE	62.77	29.93	27.66	25.65	36.42
head refit, adv. CE + smoothing 0.1	62.57	30.43	27.55	25.66	36.39
head refit, clean CE	62.70	28.90	27.23	25.15	35.90

Normalization check. Because Section 3.4 analyzes angular geometry, we check whether discarding feature magnitude changes the training conclusion. Normalizing both sides changes NRR by 0.18 with weight averaging and AWP, while raw regression is better in the base regime. We therefore retain raw squared ℓ_2 as the default and use normalized features only for analysis; CFA does not rely on a cosine objective.

Table 15: Effect of feature normalization in the anchor loss. Normalizing both representations to the unit sphere removes their magnitudes and keeps only the angle, for which the squared Euclidean loss satisfies $\|\hat{\Phi}_s - \hat{\Phi}_t\|_2^2 = 2 - 2 \cos \angle(\Phi_s, \Phi_t)$ and is therefore equivalent to cosine loss up to an affine transformation. CIFAR-100, ResNet-18, 100 epochs; within each panel, the runs differ only in how the student and teacher features are normalized. The asymmetric row normalizes the student alone, placing the two sides at different scales, and is included only as a diagnostic.

	Clean	AA	NRR
<i>(a) With the stack: WA, AWP, $\epsilon_{\text{train}} = 8.8/255$</i>			
Raw features (squared ℓ_2)	60.21	28.14	38.35
Both normalized (cosine)	61.41	28.07	38.53
<i>(b) Base regime: no WA, no AWP, $\epsilon_{\text{train}} = 8/255$</i>			
Raw features (squared ℓ_2)	62.40	24.34	35.02
Both normalized (cosine)	61.52	22.90	33.38
Asymmetric: student normalized, teacher raw	61.28	22.98	33.43

Table 16: ϵ_{train} swept with everything else fixed. Evaluation is at $\epsilon = 8/255$ in every cell.

	Clean	PGD-20	CW	AA	NRR
<i>CIFAR-100</i>					
6/255	68.07	28.40	28.16	25.82	37.44
7/255	65.61	30.34	29.64	27.44	38.70
8/255	63.89	31.41	29.83	27.78	38.72
8.8/255 [†]	62.17	32.37	30.93	28.86	39.42
10/255 [†]	59.99	32.82	30.53	28.77	38.89
<i>CIFAR-10</i>					
6/255	89.64	49.29	50.20	47.88	62.42
7/255	88.30	51.43	52.64	50.27	64.07
8/255 [†]	87.22	52.43	53.55	51.15	64.48
8.8/255 [†]	84.96	52.98	53.94	51.74	64.31
10/255 [†]	83.29	53.25	53.88	51.79	63.87

Table 17: Effect of AWP on top of weight averaging across datasets and training radii. Every run uses the sensitivity-matched radius for 100 epochs; the AWP proxy uses $\gamma = 0.005$ after a 10-epoch warmup. Within each row, AWP is the only added component.

	WA			WA + AWP		
	Clean	AA	NRR	Clean	AA	NRR
<i>CIFAR-100</i>						
8/255	63.54	26.75	37.65	63.89	27.78	38.72
8.8/255	62.30	27.69	38.34	62.17	28.86	39.42
<i>CIFAR-10</i>						
8/255	86.67	49.20	62.77	87.22	51.15	64.48
8.8/255	85.16	50.15	63.13	84.96	51.74	64.31

Table 18: Published adversarial-distillation objectives supplied with our natural teacher. The first row of each method uses its published recipe; indented rows add our teacher initialization and then without weight averaging or AWP.

	CIFAR-100			CIFAR-10		
	Clean	AA	NRR	Clean	AA	NRR
ARD	57.61	20.24	29.96	84.58	43.75	57.67
+ teacher initialization	57.96	20.22	29.98	—	—	—
RSLAD	59.68	21.30	31.40	85.15	42.91	57.06
+ teacher initialization	59.87	20.71	30.77	—	—	—
AdaAD	59.79	23.19	33.42	85.39	46.22	59.98
+ teacher initialization	59.75	22.78	32.98	—	—	—
AdaAD + IGDM	48.25	19.48	27.75	85.01	45.50	59.27
+ teacher initialization	49.67	20.15	28.67	—	—	—
PGD-AT, teacher initialization	57.73	26.46	36.29	82.23	50.72	62.74
CFA, no WA/AWP	61.29	25.36	35.88	—	—	—

Table 19: RPAT++ given the rest of our stack. Its own recipe already carries weight averaging (EMA 0.999 from epoch 105 of 200), so the missing component is the AWP proxy, added at $\gamma = 0.005$ with warmup at 10% of the run and evaluated on its own loss; the last row also starts the run at our natural teacher. CIFAR-100, ResNet-18, our reproduction, reported at both the final and the PGD-20-selected weight-averaged checkpoint.

	Final			Best by PGD-20		
	Clean	AA	NRR	Clean	AA	NRR
RPAT++ (its own recipe)	57.05	26.61	36.30	55.93	27.37	36.74
+ AWP	57.48	28.05	37.70	57.50	27.92	37.60
+ AWP + teacher initialization	57.83	27.47	37.25	57.80	27.59	37.36

E THE ALLOCATION RULE AND ITS BUDGET SOLVE

E.1 WHERE THE RULE COMES FROM

The objective is optimized over a **pixel** ball, while what it measures is a displacement in feature space, and a fixed pixel radius moves that displacement by very different amounts across samples. The allocation follows from one elementary fact: for a differentiable loss the first-order movement available inside an ℓ_∞ ball of radius e is

$$\max_{\|\delta\|_\infty \leq e} \langle \nabla_x \mathcal{L}, \delta \rangle = e \|\nabla_x \mathcal{L}\|_1, \quad (14)$$

the dual norm of ℓ_∞ being ℓ_1 . Equalizing that movement across a batch subject to a fixed total budget $\sum_i \epsilon_i = N\epsilon$ gives $\epsilon_i g_i = \text{const}$, that is $\epsilon_i \propto 1/g_i$ with $g_i = \|\nabla_x \mathcal{L}(x_i)\|_1$. The same allocation is the max-min one: if two samples move by different amounts, shifting budget from the sample that moves more to the sample that moves less raises the minimum, so no unequal assignment is optimal. Taking $\epsilon_i \propto (\bar{g}/g_i)^p$ makes $p = 1$ that allocation, $p = 0$ the uniform-pixel one, and $p \in (0, 1)$ an interpolation.

Two approximations are worth declaring rather than leaving to be found. First, g_i is measured once, by a single backward pass at the *clean* x_i before the attack starts, so it is the sensitivity at the starting point and not along the PGD trajectory. Second, everything above is first order in ϵ . Neither is repaired anywhere in the method; the rule is a budget heuristic with a derivation, not an exactly solved subproblem.

We also do not claim the rule is specific to a directional objective. $\|\nabla_x \mathcal{L}\|_1$ is defined for any loss and the implementation reads whichever loss is configured; on the shipped raw- ℓ_2 anchor it equalizes a feature displacement rather than an angle, which is why the honest name is *sensitivity-matched* ϵ rather than an angular budget. What separates it from IAAT, MMA and CAT is not directionality but the signal: they allocate from a property of the sample, this allocates from the geometry the training loss is written in.

The separation is not one of dispersion, which is worth stating because a rule that simply spread the budget more aggressively would be the simpler explanation of any gain. Measured over training on the shipped configuration, the coefficient of variation of our sensitivity signal has median 0.56, against 0.75 for the difficulty score IAAT and CAT allocate from and 1.33 for the logit margin of MMA. The box of this section compresses all three before they reach the attack, leaving per-sample multipliers of spread 0.32, 0.39 and 0.42 respectively, and the ordering survives the compression. Ours is the least dispersed of the three signals either way, so the comparison in Table 6 is a comparison of which geometry the budget follows.

One discrepancy is ours to declare rather than the reader’s to find. Equation (14) gives the ℓ_1 norm and the implementation reads ℓ_2 , for no better reason than that the cell which fixed the recipe used it and every logged run since had to stay reproducible against that choice. The two are not the same allocation. Measured over training on the shipped configuration, the ℓ_1 signal’s coefficient of variation has median 0.68 against the ℓ_2 signal’s 0.56, and the per-sample multiplier the attack receives has spread 0.34 against 0.32 — so the norms really do distribute the budget differently.

They nonetheless land close together. Training the shipped recipe with `featdir_angepts_gnorm` set to 1 and nothing else changed gives 62.16/31.97/30.56/28.57 (NRR 39.15) against 62.17/32.37/30.93/28.86 (NRR 39.42): clean accuracy is identical to a hundredth and AutoAttack differs by 0.29, in favour of the norm we ship.

We do not call that a tie. It is twice the seed-to-seed spread of Appendix F, and it is the same order of magnitude as what the allocation buys in the first place: against the uniform radius, $p = 1$ gains +1.75 clean and +0.44 AutoAttack at ℓ_2 and +1.74 and +0.15 at ℓ_1 . So the rule’s benefit survives the norm in direction and almost entirely on the clean axis, while most of its AutoAttack half does not. The honest statement is that the norm is a real choice which we made before we could compare, that the comparison favours what we ship, and that a reader preferring the derived ℓ_1 would still get the clean gain and a third of the robust one.

The permuted row of Table 6 deserves its construction spelled out, since it is what makes the comparison a comparison of signals rather than of budgets. It takes the multiset of weights the gradient rule produces — the same values, the same clip $[0.5, 1.5]$, the same restored mean — and only permutes which example receives which, so that the larger radii go to the easier examples, which is the direction IAAT and CAT assign them. Reordering costs 0.80 AutoAttack against the rule it reorders and lands below even the uniform row on AutoAttack, CW and NRR alike.

E.2 CLIPPING AND RESTORING THE MEAN RADIUS

The reported configuration floors the gradient norm at 10^{-12} and computes $v_i = \text{clip}((\bar{g}/g_i)^p, 0.5, 1.5)$, followed by $w_i = Nv_i / \sum_j v_j$ and $\epsilon_i = \epsilon w_i$. PGD uses step size $a_i = aw_i$. This preserves the batch mean radius exactly. The clipping thresholds apply before normalization, so they are not hard bounds on the final multipliers. When all gradients vanish, the floor makes all g_i equal and the allocation uniform.

Optional exact-box variant. The implementation also supports an alternative that constrains the final multipliers. It is not enabled in the reported default configuration.

With $r_i = (\bar{g}/g_i)^p$ and box $[w_{\text{lo}}, w_{\text{hi}}] = [0.5, 1.5]$, the radii that satisfy both the box and the budget exactly are $\epsilon_i = \epsilon \cdot \text{clip}(tr_i, w_{\text{lo}}, w_{\text{hi}})$ with t the solution of

$$\sum_{i=1}^N \text{clip}(tr_i, w_{\text{lo}}, w_{\text{hi}}) = N. \quad (15)$$

The left side is continuous and nondecreasing in t from Nw_{lo} to Nw_{hi} , so a root exists and bisection finds it; when no clip is active it reduces to $t = N/\sum_i r_i$, that is, to restoring the mean. Clipping first and rescaling afterwards, the obvious alternative, does not preserve the box—the rescaling moves the clipped entries back out of it.

Table 20: WideResNet-34-10 on CIFAR-100 at $\epsilon = 8/255$, measured in our framework; the CIFAR-10 column of this network is in Table 3. The CFA entry is a provisional reading of a run whose evaluation is still finishing.

Method	T.	Clean	AA	Avg	NRR
PGD-AT	–	58.97	22.84	40.91	32.93
TRADES	–	57.35	25.60	41.48	35.40
MART	–	57.16	22.43	39.80	32.22
ARD	nat.	59.52	23.05	41.29	33.25
RSLAD	nat.	60.48	21.63	41.06	31.87
AdaAD	nat.	61.66	25.62	43.64	36.20
LBGAT	nat.	59.63	28.55	44.09	38.61
ARREST	nat.	67.69	24.69	46.19	36.18
HAT	nat.	62.81	26.91	44.86	37.68
ADR	self	64.50	29.04	46.77	40.05
CFA (ours)	nat.	65.00	30.00	47.50	41.05

F EXPERIMENTAL DETAILS

Table 21: Natural teacher checkpoints used by the shipped models. The teachers are trained only on clean examples and have zero adversarial accuracy.

Dataset	Architecture	Epochs	Clean
CIFAR-10	ResNet-18	200	95.32
CIFAR-100	ResNet-18	200	77.66
Tiny-ImageNet	ResNet-18	200	66.29

Every number we report is the **final** epoch, never a checkpoint selected on a robustness metric. This matters when reading Table 28: RPAT selects on PGD-20 and ADR on PGD-10, so those rows are upper envelopes of a training run where ours is its endpoint.

Teacher. One naturally trained network per dataset, no adversarial examples anywhere in its training, and the student is warm-started from the same checkpoint. ResNet-18, SGD at learning rate 0.1 under a one-cycle schedule, batch size 128, no augmentation beyond the standard crop and flip, for 200 epochs on each dataset. The resulting teachers are summarized in Table 21. The teacher’s own robustness is 0 — it is never attacked and is not meant to survive attack; Figure 3 is the ablation over how long it is trained.

Student. The main ResNet-18 configurations are listed below. Unless noted otherwise, CIFAR-100 ablations use 100 epochs at $8.8/255$, with a 10-epoch AWP warmup.

	CIFAR-10	CIFAR-100	Tiny-ImageNet-200
Backbone	ResNet-18	ResNet-18	ResNet-18
Optimizer	AdamW	AdamW	AdamW
Learning rate / schedule	0.021, one-cycle	0.021, one-cycle	0.021, one-cycle
Epochs	100	150	100
Batch size	128	128	128
Training radius ϵ_{train}	8.8/255	8/255	8.8/255
Inner attack	PGD-10, step 2/255	PGD-10, step 2/255	PGD-10, step 2/255
Allocation exponent p	1	1	1
Radius box $[w_{\text{lo}}, w_{\text{hi}}]$	[0.5, 1.5]	[0.5, 1.5]	[0.5, 1.5]
Classifier head	frozen	frozen	frozen
Weight averaging (start, decay)	0.2, 0.999	0.2, 0.999	0.2, 0.999
AWP (γ , warmup)	0.005, 10 ep	0.005, 15 ep	0.005, 10 ep
Teacher checkpoint	clean, 200 ep	clean, 200 ep	clean, 200 ep

The head being frozen is what removes τ and β from the method: with the classifier inherited and never updated, the head-KD term of earlier drafts has no parameters to reach and is switched off entirely. p is not tuned. The training-radius sweep is reported in Table 16; evaluation is at 8/255 in every cell.

Evaluation. We report the APGD-CE/APGD-T cascade at ℓ_∞ radius $\epsilon = 8/255$ on the full test set. A dedicated validation of the CIFAR-100 seed-1 model at 100 epochs uses all four attacks in standard AutoAttack; FAB-T and Square leave the two-attack result unchanged (Appendix I). PGD-20 and CW_∞ are also reported (Appendix K), and NRR is the harmonic mean of clean and robust accuracy. Where a table reports a method we ported rather than quoted, the port uses its published recipe unless stated otherwise. The CIFAR-100 ARREST row in HAT and LBGAT retain their published schedules because both collapse under ours.

Repeated training. Each reported row is an individual run. For the 100-epoch CIFAR-100 configuration at 8.8/255, seed 1 gives 62.00/32.49/30.82/28.74 clean/PGD-20/CW/AA, compared with 62.17/32.37/30.93/28.86 for seed 0. These two runs provide a limited reproducibility check, not a variance estimate or a significance threshold for other comparisons. In particular, small differences between regression targets or nearby operating points require further repetitions.

Compute, teacher included. A single RTX 5080. One student cell is about 2.2 hours on CIFAR-100 at 100 epochs, of which the AutoAttack evaluation is roughly 7 minutes; Tiny-ImageNet is longer in proportion to its test set.

The teacher has to be counted, since calling it free would otherwise hide a training run. It is not free, but it is cheap: a clean epoch has no inner maximization, so it costs 6.8 seconds against 74.5 for an adversarial one, and 200 clean epochs come to 0.39 hours. The teacher is therefore 19% of one student run and is amortized over every cell that reads it, the teacher ladder of Figure 3 included. Table 22 puts the total against the baselines measured on the same machine.

Table 22: Wall-clock training cost on one RTX 5080, CIFAR-100, ResNet-18, as logged by the runs reported in this paper. Evaluation is excluded. Adversarial distillation additionally requires an adversarially trained teacher, which is not counted in its row and is the more expensive artifact.

Method	Epochs	Teacher (h)	Student (h)	Total (h)
Natural teacher (ours, reused)	200 clean	–	0.39	0.39
CFA (ours)	100 adv.	0.39	2.07	2.46
Label CE at our regime	100 adv.	0.39	3.22	3.61
PGD-AT, initialized at the teacher	100 adv.	0.39	1.72	2.11
ARD	100 adv.	0.39	1.52	1.91
RSLAD	100 adv.	0.39	1.71	2.10
AdaAD	100 adv.	0.39	2.79	3.18
LBGAT	100 adv.	joint	1.88	1.88
HAT	50 adv.	0.39	1.10	1.49
ADR	200 adv.	self	3.52	3.52

Two readings follow. The anchor is not bought with compute, since it costs less end to end than label cross-entropy at the same regime and about what PGD-AT costs once the teacher is included. And the comparison with adversarial distillation is favourable in a way the table understates, because those rows assume a robust teacher that the current literature obtains from a WideResNet trained on generated data, which is more expensive than every row here combined.

G TEACHER-ONLY METRICS AND WHAT THEY PREDICT

Section 3.4 selects a teacher by training students. This appendix asks what can be read off the frozen teacher instead, reports the predictions that failed as well as the one that held, and records a measurement we retracted. Metrics are computed on 5000 held-out images: clean accuracy, the scatter ratio S_w/S_b , the mean logit margin $f_y - \max_{k \neq y} f_k$, the mean $\|\nabla_x \text{CE}\|_2$, and the feature sensitivity $\|\Phi_t(x + \delta) - \Phi_t(x)\|_2 / \|\delta\|_2$ for uniform δ at 8/255. The students are the 50-epoch cells of Table 11, identical apart from the teacher.

Table 23: Left: Spearman correlation between a teacher-only metric and the student it produced, over the nine trajectory checkpoints. Right: Spearman between the metrics themselves, which is what makes the left block hard to read — teacher clean accuracy has correlation 1.00 with the epoch index, so any quantity monotone in training time inherits its correlations.

Teacher metric	Student			Teacher metric		
	Clean	AA ₂	NRR	Epoch	Clean	Margin
Clean accuracy	0.18	0.95	0.95	1.00	1.00	0.90
S_w/S_b	-0.17	-0.93	-0.93	-0.98	-0.98	-0.88
Logit margin	0.30	0.98	0.98	0.90	0.90	1.00
$\ \nabla_x \text{CE}\ $	0.90	0.48	0.48	0.43	0.43	0.53
Feature sensitivity	0.93	0.40	0.40	0.32	0.32	0.47
Margin / sensitivity	0.18	0.95	0.95	1.00	1.00	0.90

Two diagnostics temper the left block. Leave-one-checkpoint-out, fitting a line on eight checkpoints and predicting the ninth, the student’s AutoAttack accuracy is predictable to 0.56 points from the margin and 0.78 from teacher accuracy, against 2.47 for predicting the mean; the student’s clean accuracy is much harder, 1.32 points from the gradient norm and 1.72 from teacher accuracy against 1.90 for the mean. And regressing the student on teacher accuracy leaves residuals whose rank correlation with the margin is 0.48 and with feature sensitivity 0.30, neither of which $n = 9$ supports. The trajectory therefore selects candidates; it does not test them.

Table 24: Out-of-sample prediction: each line is fit on the nine trajectory checkpoints and applied to the three teachers of Table 1, which were trained differently. Mean absolute error in accuracy points, and the error on each teacher.

Predicted	From	MAE	Label sm.	Mixup	Weight decay
Student clean	Feature sensitivity	2.99	0.7	1.2	7.2
Student clean	$\ \nabla_x \text{CE}\ $	3.51	2.2	0.1	8.3
Student clean	Teacher clean accuracy	3.45	4.7	0.4	5.2
Student AA ₂	Logit margin	3.05	3.3	5.5	0.4
Student AA ₂	S_w/S_b	7.95	2.6	12.0	9.3
Student AA ₂	Teacher clean accuracy	0.92	0.4	0.4	2.0

The ordering reverses between the two tables. The margin and the scatter ratio, strongest along the trajectory, are the weakest predictors off it: the margin predicts 22.5 and 20.2 AutoAttack points for the label-smoothing and mixup teachers, which give 25.72 and 25.70, and the scatter ratio errs by 12.0 points on the mixup teacher. Their trajectory correlations were the collinearity with training time. Teacher accuracy predicts the robust axis best out of sample and the clean axis worst, and feature sensitivity is the reverse. Neither the label-dependent gradient norm nor the margin is the right quantity to read on a label-smoothing teacher in any case, since label smoothing changes the logit scale both of them are computed from, which is a second reason to name the feature version in Section 3.4.

One measurement was retracted rather than reported. A cosine-schedule checkpoint was to be a fourth off-trajectory teacher, and its metrics looked decisive: a margin of 0.22 at 73.00 clean accuracy. It is the normalized-feature variant of this network, carrying a feature-scale parameter the plain network does not, and the metric script had loaded it with that parameter dropped. Its student could not be trained at all, which is how the error surfaced. The script now refuses a checkpoint whose parameters do not match the network, and the three teachers of Table 1 use the same network as the ladder.

H THE RECIPE CONTROLS

Unless explicitly ablated, CFA warm-starts from the natural teacher and trains with AdamW at 0.021 under a one-cycle schedule, while the baselines retain their authors’ recipes. Table 25 asks whether that recipe is what wins: it runs our own objective, and label cross-entropy for reference, under both recipes and both initializations. Every row is measured here at the base regime, which carries no weight averaging, no AWP and a uniform radius, on CIFAR-100 with ResNet-18 for 100 epochs.

Table 25: The recipe controls. *Ours* is AdamW at 0.021 under a one-cycle schedule with the teacher warm start; *theirs* is SGD at 0.1 decayed by ten at epochs 70 and 90, the schedule PGD-AT uses here. The training radius and every other field are held fixed within each pair.

Objective	Recipe	Clean	AA	Avg	NRR
Anchor	ours, teacher init	61.29	25.36	43.33	35.88
Anchor	ours, random init	60.81	25.21	43.01	35.66
Anchor	theirs, teacher init	59.65	25.54	42.60	35.77
Label cross-entropy	ours, teacher init	54.60	19.45	37.03	28.68
Label cross-entropy	theirs, teacher init	56.66	20.91	38.79	30.55

Changing initialization or schedule changes the anchor’s NRR by 0.22 and 0.11, respectively. Cross-entropy improves by 1.87 NRR under its own schedule, so Table 4 includes that stronger baseline. The anchor remains ahead under both tested schedules. These comparisons support an objective-level contribution within the tested recipes, without establishing significance from single runs or claiming exhaustive tuning of the alternatives.

I GRADIENT MASKING

We validate the reported robustness with the full standard AutoAttack suite and additional gradient-masking diagnostics (Athalye et al., 2018). The evaluated model is the CIFAR-100 seed-1 repeat of the 100-epoch configuration at 8.8/255, with 62.00% clean accuracy and 28.74% in its original two-attack evaluation. In the dedicated validation on all 10000 test images, both the two-attack cascade and the full four-attack suite yield 28.73% (Table 26). This establishes that FAB-T and Square do not reduce robust accuracy further for this checkpoint; it does not substitute for a four-attack evaluation of other checkpoints.

Square leaves 5.99 points more accuracy than the strongest individual white-box attack, APGD-T. In the standard cascade, neither FAB-T nor Square successfully perturbs any of the 2873 examples that survive APGD-CE and APGD-T, leaving robust accuracy unchanged at 28.73%. PGD accuracy decreases monotonically over the tested radii, reaching zero at 64/255, and decreases with the number of steps, with smaller changes beyond 50. Together, these checks provide no indication of gradient masking under the evaluated attacks, without establishing worst-case robustness.

The transfer row is worth reading separately. The student inherits the teacher’s classifier and is initialized at the teacher’s features, so adversarial examples built on the teacher might be expected to carry over; they cost the student 1.58 points against its clean accuracy. Sharing a head is therefore not sharing a vulnerability, which is the same conclusion Section 5.4 reaches from class separation: what the anchored student changes is the position of its features, and the teacher’s own adversarial directions do not point at the student’s boundaries.

Table 26: Gradient-masking checks, all 10000 test images, $\epsilon = 8/255$. The first block runs each attack *independently* from the clean test set, which is what compares their strengths; the second runs them as AutoAttack does, each attack seeing only the points the previous ones left standing. The teacher-transfer row builds PGD-20 examples on the natural teacher and evaluates them on the student, which shares the teacher’s classifier.

Check	Setting	Accuracy
<i>white box against black box</i>		
APGD-CE	white box, untargeted	31.80
APGD-T	white box, targeted	28.83
FAB-T	white box, boundary	29.16
Square	black box , 5000 queries	34.82
<i>as a cascade</i>		
APGD-CE $\rightarrow$ APGD-T	two-attack validation	28.73
$\rightarrow$ FAB-T $\rightarrow$ Square	AutoAttack’s standard version	28.73
<i>radius sweep, PGD-20</i>		
$\epsilon = 2/255$		54.62
$\epsilon = 4/255$		47.03
$\epsilon = 8/255$		32.46
$\epsilon = 12/255$		20.88
$\epsilon = 16/255$		11.94
$\epsilon = 32/255$		1.05
$\epsilon = 64/255$		0.00
<i>step sweep, $\epsilon = 8/255$</i>		
PGD-10 / 20 / 50 / 100		33.17 / 32.57 / 32.30 / 32.24
<i>transfer</i>		
Natural teacher $\rightarrow$ student	PGD-20 built on the teacher	60.42

J NOTES ON THE BASELINES

Several rows of Table 2 run a specific variant of the method they name, or are grouped in a way that needs saying. This section is those decisions; the table carries none of them.

ARREST runs its published recipe on both datasets: adversarial finetuning for 20 epochs from the natural model with SGD at 0.025, the angular representation term at $\lambda = 50$, and Noisy Replay over the first half at $\phi = 30^\circ$. We read its representation after global pooling, where our anchor and every geometry measurement in this paper are defined.

LBGAT, and why its CIFAR-10 cell is empty. The authors release a training script for CIFAR-100 only, and their CIFAR-10 entries are released as pretrained WideResNet checkpoints. Running their CIFAR-100 recipe on CIFAR-10 diverges here within thirty steps: the squared error is reduced with mean over batch and classes, so a ten-class problem receives ten times the gradient of a hundred-class one at equal logit error, and $\text{lr} = 0.1$ is above the stability threshold that follows. Lowering the learning rate trains (at 0.01 the natural branch reaches 49% within 180 steps), but a learning rate we chose is not their recipe, so the cell is left empty rather than filled with a number of our own construction. LBGAT trains its natural branch jointly with the robust one, so unlike the distillation rows its teacher is not obtained for free, and we group it accordingly. The two datasets run different variants because the authors publish different ones: CIFAR-100 is LBGAT6 and CIFAR-10 is LBGAT0, their release lists CIFAR-10 checkpoints under LBGAT0 alone, and $\beta = 6$ is their CIFAR-100 setting. We report the configuration each dataset has a published counterpart for rather than forcing one β across both.

HAT uses a naturally trained network only to *label* helper points beyond the ball, never as a target, so it belongs with the trade-off methods and not the distillation ones. It is run on its own one-cycle schedule rather than ours, which collapses it to chance on both datasets — the same allowance we make for LBGAT. On CIFAR-10 the port reproduces HAT’s published no-extra-data figure to within 0.31 clean and 0.27 AutoAttack (84.90/49.08 reported, ResNet-18), which is what licenses reporting its CIFAR-100 row at all: HAT publishes no CIFAR-100 result without additional data, its

61.50/28.88 being trained with DDPM-generated samples, so that column has no published figure to be read against.

ADR is run on separate hardware and its row is filled from there. Two things have to hold when it is. ADR reports two configurations — AT+ADR at 56.10/26.87 on CIFAR-100 and AT+WA+AWP+ADR at 57.36/28.50 — and the row has to name which, the second being the one component-matched to our recipe, which carries both weight averaging and AWP. ADR is also self-distillation: its teacher is an exponential moving average of the student, no external network is read, and the student trains from scratch, so its row inherits nothing of our natural teacher.

What the published tables are blocked by. Table 28 and Table 29 separate their rows by network, and the separations are not cosmetic. RPAT’s “ResNet-18” is `ResNet(Bottleneck, [2, 2, 2, 2])` in its released code, 14.35M parameters against the 11.27M of the standard `BasicBlock` network used by us and by ADR; they also ship a `pre_resnet18` at 11.27M and use `PreActResNet-18` explicitly elsewhere, so the distinction is theirs. HAT’s Tiny-ImageNet row is `PreActResNet-18`: ADR lists it under ResNet-18, but the figures are quoted from HAT’s own Table 9, whose caption says `PreAct`, and HAT’s code refuses anything else on that dataset — same parameter count as ours, different network, and its two companions from that table are shown beside it. F²AT’s plain SAT baseline reaches 31.52/9.37 where ADR’s plain AT reaches 45.87/18.06, a gap no difference of method explains, so its rows are separated too. TE’s architecture we could not verify — it is ADR’s attribution to a paper we do not have — and it is marked accordingly. ADR trains Tiny-ImageNet for 80 epochs against our 100, which favours us and is stated here rather than left implicit. LBGAT reports that dataset without AutoAttack and B-MTARD reports it on MobileNet-v2, so neither appears; CURE, ARREST and Generalist++ do not report it at all.

IGDM is reported at $\alpha = 1$. Its published $\alpha = 20$ diverges to chance accuracy on both datasets when the teacher is natural, as documented in Table 18.

Teacher-only KL with the training stack. This diagnostic combines probability-target distillation with WA and AWP; it is not a reproduction of a published method. Temperature is applied only to the teacher, and the best tested stacked setting, $\tau = 4$, gives 56.97/26.96 clean/AA; $\tau = 16$ gives 42.45/20.10. The comparison concerns this KL training design, not all logit-based supervision. In particular, fixed-head logit MSE is evaluated separately in Table 12, where it remains competitive with feature regression.

K PER-ATTACK METRICS

Table 27 gives every attack we measure for our own cells. The main tables report clean accuracy and AutoAttack because those are the two axes the trade-off is defined on, and because PGD-20 is not a faithful reading of this method.

Why PGD-20 is not the metric here. The student inherits the teacher’s classifier and never retrains it. PGD-20 ascends the cross-entropy *through that classifier*, so it reads a head that was fitted for a different feature distribution; AutoAttack, which includes a targeted APGD, and CW_∞ , which attacks the logit margin directly, do not depend on that fit in the same way. The consequence is measurable on a controlled pair. The first two rows of Table 27 share a backbone objective, a schedule and an attack, and differ only in whether the classifier is refit:

	PGD-20	CW	AA
head-KD term retained	36.26	30.65	28.68
head frozen	32.63	30.66	28.77
difference	−3.63	+0.01	+0.09

Changing the classifier moves PGD-20 by 3.63 points and moves CW and AutoAttack by hundredths. The same pattern recurs across our ablations—the head-temperature sweep, the label-smoothing sweep, the directional-versus-raw comparison, and the radius sweep all show PGD-20

Table 27: Per-attack accuracy (%) for our cells. ResNet-18, $\epsilon = 8/255$, final-epoch weight-averaged model, full test set.

Dataset	Cell	Clean	FGSM	PGD-20	CW	AA	NRR
CIFAR-100	CFA, head frozen (shipped)	62.17	36.91	32.37	30.93	28.86	39.42
	the same cell at seed 1	62.00	36.79	32.49	30.82	28.74	39.27
	ℓ_1 gradient norm	62.16	36.48	31.97	30.56	28.57	39.15
	CFA, head-KD term retained	62.35	38.80	36.26	30.65	28.68	39.29
	CFA, $\epsilon_{tr} = 6/255$	68.07	35.88	28.40	28.16	25.82	37.44
	CFA, $\epsilon_{tr} = 7/255$	65.61	36.32	30.34	29.64	27.44	38.70
	CFA, $\epsilon_{tr} = 8/255$	63.89	36.67	31.41	29.83	27.78	38.72
	CFA, $\epsilon_{tr} = 10/255$	59.99	36.50	32.82	30.53	28.77	38.89
	CFA, no WA / AWP	61.29	33.97	28.23	27.19	25.36	35.88
	Logit anchor, our stack	56.97	35.82	33.68	28.94	26.96	36.60
CIFAR-10	RPAT++ (our reproduction)	55.93	34.46	32.44	29.37	27.36	36.74
	CFA, $\epsilon_{tr} = 8.8/255$ (shipped)	84.96	59.58	52.98	53.94	51.74	64.31
	CFA, $\epsilon_{tr} = 6/255$	89.64	59.37	49.29	50.20	47.88	62.42
	CFA, $\epsilon_{tr} = 7/255$	88.30	59.59	51.43	52.64	50.27	64.07
	CFA, $\epsilon_{tr} = 8/255$	87.22	59.92	52.43	53.55	51.15	64.48
	CFA, $\epsilon_{tr} = 10/255$	83.29	58.98	53.25	53.88	51.79	63.87
	RPAT++ (our reproduction)	82.41	59.73	55.72	52.88	50.75	62.82

moving against CW and AutoAttack. We therefore report PGD-20 for completeness and read the trade-off on clean accuracy and AutoAttack, which is also what ADR and ARREST report in their main tables.

L PUBLISHED RESULTS, FOR CONTEXT

Table 2 contains only numbers we measured ourselves. Table 28 collects the published figures for the same and neighbouring methods, so that our runs can be placed against the literature without the two being mixed in one column. These are quoted as reported and were produced by other codebases, other schedules and other checkpoint-selection rules—RPAT selects its best checkpoint by PGD-20 and ADR by PGD-10, where every number of ours is the final epoch—so they should be compared within a source rather than across.

Table 28: Published results, ℓ_∞ , $\epsilon = 8/255$, quoted as reported. **The architecture changes between blocks** and is named in each heading; only the ResNet-18 blocks are comparable with Table 2. These were produced by other codebases, schedules and checkpoint-selection rules, so they should be compared within a source rather than across.

Method	Teacher	Clean	AA	Avg	NRR	Source
<i>CIFAR-100, ResNet-18</i>						
PGD-AT	–	56.56	25.02	40.79	34.69	RPAT
TRADES	–	55.39	24.51	39.95	33.98	RPAT
MART	–	49.83	25.00	37.41	33.30	RPAT
Consistency-AT	–	58.53	25.39	41.96	35.42	RPAT
Consistency-AT + RPAT	–	60.33	26.31	43.32	36.64	RPAT
F ² AT	–	54.19	23.24	38.71	32.53	F ² AT
Generalist++	co-tr.	62.97	23.96	43.47	34.71	Generalist++
ADR (AT + ADR)	self	56.10	26.87	41.48	36.34	ADR
ADR (AT + WA + AWP + ADR)	self	57.36	28.50	42.93	38.08	ADR
<i>CIFAR-10, ResNet-18</i>						
PGD-AT	–	82.78	44.63	63.71	57.99	CURE
TRADES	–	82.41	48.37	65.39	60.96	CURE
MART	–	80.70	47.49	64.10	59.79	CURE
ST-AT	–	83.10	50.50	66.80	62.82	CURE
IAD	robust	80.63	50.17	65.40	61.85	CURE
Consistency-AT + RPAT	–	84.12	48.98	66.55	61.91	RPAT
Generalist++	co-tr.	89.09	46.07	67.58	60.73	Generalist++
ARREST	natural	86.63	46.14	66.39	60.21	ARREST
CURE	self	86.76	49.69	68.23	63.19	CURE
ADR (AT + ADR)	self	82.41	50.38	66.40	62.53	ADR
<i>CIFAR-10, PreActResNet-18</i>						
WA (Izmailov et al., 2018)	–	83.50	49.89	66.70	62.46	RPAT
AWP (Wu et al., 2020)	–	81.11	50.09	65.60	61.93	RPAT
KD + SWA (Chen et al., 2021)	robust	84.06	49.82	66.94	62.56	RPAT
TE (Dong et al., 2022)	–	82.04	50.12	66.08	62.23	RPAT
ReBAT (Wang et al., 2023)	–	82.09	50.72	66.41	62.70	RPAT
RPAT++ (Liu et al., 2025)	–	82.63	51.00	66.82	63.07	RPAT
<i>CIFAR-100, PreActResNet-18</i>						
WA (Izmailov et al., 2018)	–	57.26	25.83	41.55	35.60	RPAT
AWP (Wu et al., 2020)	–	54.10	25.16	39.63	34.35	RPAT
KD + SWA (Chen et al., 2021)	robust	57.17	25.66	41.42	35.42	RPAT
TE (Dong et al., 2022)	–	56.41	25.84	41.13	35.44	RPAT
ReBAT (Wang et al., 2023)	–	56.13	27.60	41.87	37.00	RPAT
RPAT++ (Liu et al., 2025)	–	56.84	27.68	42.26	37.23	RPAT
<i>CIFAR-10, WideResNet-34-10</i>						
WA (Izmailov et al., 2018)	–	87.66	52.65	70.16	65.79	RPAT
MMA (Ding et al., 2020)	–	87.80	43.10	65.45	57.82	RPAT
AWP (Wu et al., 2020)	–	85.63	53.32	69.48	65.72	RPAT
KD + SWA (Chen et al., 2021)	robust	87.45	53.59	70.52	66.46	RPAT
TE (Dong et al., 2022)	–	85.97	52.88	69.43	65.48	RPAT
ReBAT (Wang et al., 2023)	–	85.25	54.78	70.02	66.70	RPAT
ADR (Wu et al., 2024)	self	84.67	53.25	68.96	65.38	RPAT
CURE (Gowda et al., 2024)	self	87.05	52.10	69.58	65.19	RPAT
RPAT++ (Liu et al., 2025)	–	86.76	54.97	70.87	67.30	RPAT

Read against Table 2 and restricted to the ResNet-18 blocks, which are the ones our numbers are comparable with, our CFA row exceeds every published Avg and NRR on both datasets: 64.75/27.98 at Avg 46.37 on CIFAR-100 and 87.37/51.33 at Avg 69.35 on CIFAR-10.

The restriction is not a convenience. The WideResNet-34-10 block exceeds ours on both criteria—RPAT++ reaches Avg 70.87 and NRR 67.30 there—because a 48.3M network is a different model, not a better method, and Tables 3 and 20 are where that comparison belongs. The same caution applies to the PreActResNet-18 blocks in the other direction: RPAT++ falls to 82.63/51.00 there, and reading that as a defeat for RPAT would be the identical error. We state all of this as a comparison across codebases and architectures rather than as a controlled result, which is what Table 2 is for.

Table 29: Published Tiny-ImageNet-200 results, ResNet-18, ℓ_∞ , $\epsilon = 8/255$, quoted as reported. Our row is above all of them on both axes. The blocks are separated because the networks are not the same, and Appendix J says how each differs.

Method	Clean	AA	Avg	NRR	Source
CFA (ours)	55.16	20.54	37.85	29.93	—
AT + WA + ADR	48.55	20.23	34.39	28.56	ADR
AT + WA + AWP + ADR	48.27	20.12	34.20	28.40	ADR
TRADES + WA + AWP + ADR	51.38	19.48	35.43	28.25	ADR
TRADES + WA + ADR	51.99	19.17	35.58	28.01	ADR
AT + WA + AWP	48.61	19.58	34.09	27.92	ADR
AT + WA	49.10	19.30	34.20	27.71	ADR
TE*	47.46	18.29	32.88	26.40	ADR
AT	45.87	18.06	31.96	25.92	ADR
TRADES	48.49	17.35	32.92	25.56	ADR
<i>PreActResNet-18 — same parameter count, different network</i>					
HAT	52.60	18.14	35.37	26.98	HAT
TRADES	48.25	17.17	32.71	25.35	HAT
AT	47.76	17.92	32.84	26.02	HAT
<i>Bottleneck ResNet-18 (14.35M) and an unstated setup — see caption</i>					
Consistency-AT + RPAT	49.74	18.84	34.29	27.33	RPAT
PGD-AT + RPAT	47.68	17.77	32.73	25.89	RPAT
Consistency-AT	46.54	17.60	32.07	25.54	RPAT
MART	39.70	17.18	28.44	23.98	RPAT
F ² AT	40.54	13.13	26.84	19.84	F ² AT
SEAT	35.51	11.37	23.44	17.22	F ² AT
SAT	31.52	9.37	20.45	14.45	F ² AT

M THE CURE REPRODUCTION

Seven runs of the official CURE repository, from its own code and our shared CIFAR data, none of which reproduce the published 86.76/49.69 on CIFAR-10. They fall short on *different* axes: the closest on standard accuracy reaches 86.11 at AA 40.65, the closest on robustness AA 45.63 at clean 81.15, while the published pair has both. Running the repository unmodified gives 65.03/29.74. Computing its RGP prominence mask on absolute values raises AA by 15.9 points (29.74 $\rightarrow$ 45.63), which localizes the discrepancy to that step without closing it.

We re-evaluated all five surviving checkpoints with our own pipeline and it agrees with the repository’s own evaluation to within 0.05 AutoAttack on every one, so the gap is in training rather than in measurement. For contrast, RPAT++ reproduces to within 0.31 AutoAttack of its published numbers on both datasets under the same conditions. The CURE row of Table 28 is therefore the paper’s own number, and CURE is omitted from Table 2.

N A CORRECTION TO THE AWP PROXY, AND WHAT IT MOVED

AWP ascends an auxiliary objective in weight space before each descent step. In our implementation the closure used for that ascent had drifted from the closure used for descent, in two ways. It

retained a head-distillation term that `freeze_head` removes from training, so AWP was maximizing a quantity partly composed of a term the model never optimizes; and a detach intended for the normalized feature was applied on a path that feeds the unnormalized one. Measured on CIFAR-100 before the fix, the spurious term was 7.15 against the anchor’s 4.15 and contributed 40.8% of the AWP backbone gradient, at cosine 0.918 to the correct direction. Separately, the head was frozen *after* the proxy was copied, which is harmless only because the proxy’s first step is delayed by the AWP warmup.

We report this because it is the kind of defect that changes a paper’s numbers, and because here it did not. Re-running the shipped recipe with both closures agreeing gives 62.17/28.86 against 62.65/28.77 on CIFAR-100 and 84.96/51.74 against 85.58/51.79 on CIFAR-10: NRR moves by 0.01 and 0.22, no ordering in any table changes, and clean accuracy moves more than robustness does. AWP is evidently insensitive to a large, well-aligned perturbation of its ascent direction, which is consistent with its role here as a regularizer of sharpness rather than a precisely targeted objective. All numbers reported for the shipped recipe are post-correction. Two exceptions are stated where they appear: the head-freeze comparison in Appendix K and the ϵ_{train} sweep in Table 16 were measured before it. In each, every arm shares the defect, so the comparison is internally consistent even though its absolute values differ from the corrected ones by a few hundredths.

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